

The Impact of the Course on Putting Within 10 Feet

Group 7

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1. Introduction to Data

In golf, there are multiple ways a player can improve their game, but none are more important than short-range putts. These short-range putts consist of those 10 feet and closer, which ultimately prove to be the most crucial to a player's score and tournament outcome. Professional players are expected to convert these at an extremely high rate, approaching 88%; however, even within 10 feet, these outcomes can vary significantly. While casual players view these putts as straightforward, in reality, their success hinges on a complex combination of factors, including green conditions, slope, pressure situations, and individual player tendencies. Understanding these factors is not only relevant for performance analysis but can also be crucial to optimizing decision-making on the course and statistically improving how the game is played.

The goal of this project is to conduct a statistical investigation into what factors most strongly affect the likelihood of a golfer converting a putt within 10 feet. Rather than relying on broadcaster commentary or just pure intuition, we wanted to pursue a data-driven approach, observing tournament scenarios and quantifying the player-level elements for each attempt. Our research question is therefore “which factors most strongly influence whether a golfer successfully makes a putt from within ten feet?” Due to this being a statistical inquiry, we based our hypothesis around a formal hypothesis structure:

- **Null Hypothesis (H_0):** None of the observed variables, such as distance, break direction, slope severity, hole difficulty, pressure situation, green speed, or player performance statistics, has a significant effect on whether a golfer makes a putt inside ten feet.
- **Alternative Hypothesis (H_1):** At least one of these variables has a significant effect on the probability of making a putt inside ten feet.

Name	Putt Made	Distance (Ft)	Break Direction	Slope	Green Speed	Hole Difficulty	Round	Pressure Situation	TOB(s)	Avg	Make %
Scottie Scheffler	Yes	4-6	Straight	Slight	13	Medium	4	Last Round	10	1.68	88.89%
Patrick Cantlay	Yes	4-6	L-R	Slight	13	Medium	3	Last Round	13	1.74	83.6%
Tommy Fleetwood	Yes	0-3	L-R	Steep	13	Hard	2	Final-holes	9	1.76	83.5%
Viktor Hovland	Yes	0-3	R-L	Slight	13	Hard	2	Last Round	15	1.78	73.9%
Tommy Fleetwood	Yes	0-3	Straight	Flat	13	Medium	4	Normal	13	1.76	83.5%

Table 1 – Data Preview (note: some columns are represented by abbreviations)

When collecting our data, we looked at a total of eleven variables. Each of the variables we collected focused on golf at the putt level. Along with each observation row, we added a column

to detail the name of the golfer we were observing. Each row of variables in our dataset represents the first putt made at a hole. Golfers were allowed to repeat in the same tournament, so long as the hole that they were playing at was different. These variables were collected by watching various past PGA tournaments (years 2021-2025) and focusing only on the putts. We also used other tools, such as TourCast and the PGA Tour website, to get various details that may not have been stated during the broadcast. Several of the observations made had some variables that could not be found. After removing observations with N/A values, we collected 189 observations in total. From the observational variables that we collected, seven of the eleven variables were categorical.

- **Putt made:** This variable was used to signify whether or not a putt was successfully made during the observation. These were noted with either a “yes” or “no.” To get this variable, we watched the first putt of the hole. If the ball went into the hole, it was marked as “yes.” If the first putt missed, the observation was marked as “no.”
- **Distance in feet:** This variable was used to note how far the golfer had to putt to make it into the hole. We split the distances into three buckets: 0-3 feet, 4-6 feet, and 7-10 feet. The distance from the hole was given to us either from the broadcast, where the announcers would state how far the ball was from the hole, or it was taken from TourCast. We were able to get the distance from TourCast by matching the round number and hole number from the broadcast. From there, the tool would give us the exact distance the ball was from the hole at the first putt.
- **Break Direction:** Break direction is used to signify which direction the golfer had to aim to make a successful putt. We split this variable up into three different directions: left to right (L-R), right to left (R-L), and straight. This variable could only be recorded by watching the direction in which the ball moved from the broadcast. In the case of an L-R or R-L, we recorded the direction based on the perspective of the golfer and where they would have needed to hit the ball.
- **Slope severity:** Slope severity is the shape of the slope from where the ball landed to the hole. These were categorized into three different classifiers: flat, slight, and steep. The results from this variable were collected from the broadcasts by looking at where the ball landed in relation to the hole and how much of an incline there was between the ball and the hole.
- **Hole difficulty:** Hole difficulty was used to identify which holes were easier or harder. This was split up into three categories: easy, medium, and hard. To collect the data for this variable, we collected it in two ways. The first way was by listening to the announcer when they are introducing the hole. The second, if the announcer did not mention the difficulty, was to look at the PGA Tour’s website. On the course description, they ranked the hole difficulty from 1-18 (1 being the hardest). From there, we divided the difficulty ranking by three so that 1-6 was hard, 7-12 was medium, and 13-18 was easy.
- **Pressure situation:** Pressure situation was used to identify where in the tournament the golfer was during our observation. The pressure situation was divided into four different

categories: normal, cut-line, final holes, and last round. This was recorded by looking at which round the observation was taking place in. Rounds 1 and 3 generally had a pressure category of normal. For these two rounds, if the golfer was on their last holes for the round, instead of giving them the normal pressure category, we would give them the final holes pressure category. The threshold for final holes would be any hole after the 10th hole until the 18th. Round 2 generally had a pressure classification of cut-line, and Round 4 was classified as the final round.

- **Round number:** Round number was an ordinal variable that we used to supplement a few of the variables (such as pressure situation). This variable was taken by looking at where the broadcast was in the tournament. This could usually be found in the title of the broadcast or will be announced at the beginning of the broadcast.

The remaining four variables were numerical.

- **Green speed:** The green speed, or the stimp, was used to find out how fast the ball would go on the particular course that the golfer was playing on. This was found by using the PGA Tour's website to give the stats for the course.
- **Time spent over the ball in seconds (TOB(s)):** Time spent over the ball is the number of seconds that the golfer spends before making their putt. This was collected by timing the golfer with a stopwatch from the moment they start standing over the ball to when they start their swing. The time gained from the stopwatch was rounded to the nearest second to account for the possible error of starting/stopping.
- **Average putts per hole (Avg):** The average putts per hole is the golfer's average number of times that they will putt during the season of the tournament during which the observation was recorded. For example, a golfer would have their average putts per hole recorded from the 2022 season if the tournament was played during the 2022 season. This information was taken from the PGA Tour's website for player stats.
- **Golfers' make % within 10 feet (Make %):** This variable measures the success percentage of a golfer's putts with the ball being within 10 feet of the hole. Similar to average putts per hole, this variable was recorded based on the season in which the tournament was played, taking from the PGA Tour's website for player stats.

2. Summary of the Data

Variable Title	Min	Max	Mean	SD
Green Speed	10.5	13	12.65	0.75
TOB(s)	1	16	7.43	3.83
Average Putts Per Hole	1.57	1.82	1.73	0.04
Make % within 10 ft.	32.35	89.98	82.93	12.79

Table 2 – Numeric Variables

Table 2 summarizes the four numeric variables, giving us the lowest values in each variable (min), the highest values (max), the average (mean), and the standard deviation (SD).

Variable Title	Category	Count	Percentage % (rounded to the 10th)
Putt Made	Yes	150	79.4%
	No	39	20.6%
Distance (feet)	0-3	87	46%
	4-6	56	29.6%
	7-10	46	24.3%
Slope Severity	Flat	114	60.3%
	Slight	60	31.7%
	Steep	15	7.9%
Hole Difficulty	Easy	78	41.3%
	Medium	47	24.9%
	Hard	64	33.9%
Pressure Situation	Normal	44	23.3%
	Cut-line	5	2.6%
	Last Holes	27	14.3%
	Final Round	113	59.8%
Round	1	32	16.9%
	2	20	10.6%
	3	31	16.4%
	4	106	56.1%

Table 3 – Categorical Variables

Table 3 summarizes the seven categorical data points that we collected. Each row shows subrows that contain the total amount of recorded observations for the categories and the percentage that each category makes up for the variable.

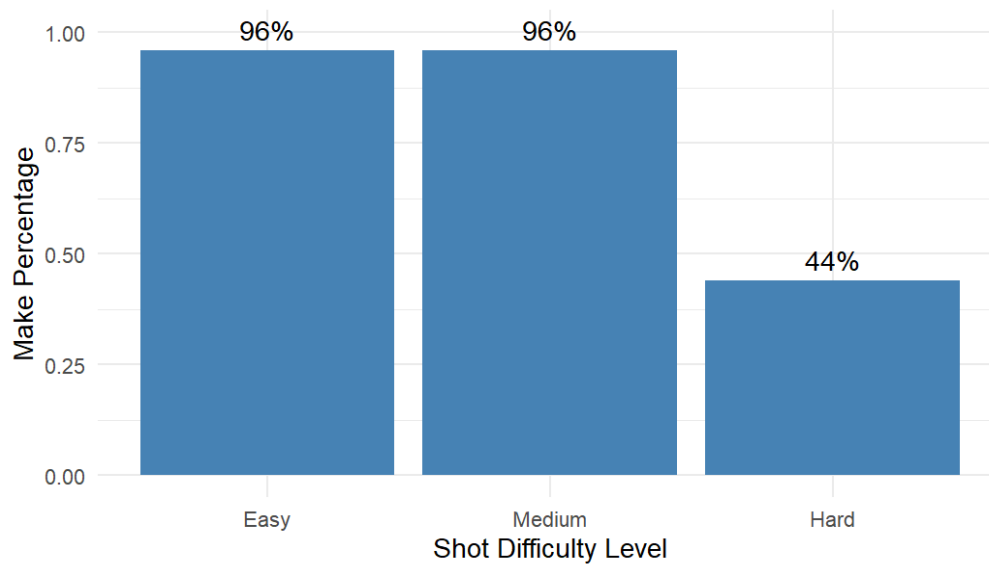


Figure 1 – Success Rate vs. Shot Difficulty

Figure 1 shows us the relationship between the putt's success percentage and the difficulty of the shot that was being made. Shot difficulty was a variable created using the variables that we observed and held various weights. A few of the variables (distance, break direction, slope, and hole difficulty) were categorical, so we converted them into values from 1 through three based on difficulty. Lower distances (0-3) were given scores of 1, while higher distances (7-10) were given scores of 3. Similar methodologies were given to the slope variables. For the break direction, we gave straight the lowest value, and R-L the highest value. This is due to the average handedness of golfers. The weights were determined based on the subjective importance of what they add to a putt shot. Distance was given the highest weight of 3. Break direction and slope were given the next highest weight of 2. Green was given a weight of 1.5. Hole difficulty was given a weight of 1.

$$\text{Shot Difficulty} = 3(\text{distance}) + 2(\text{break}) + 2(\text{slope}) + 1.5(\text{green}) + \text{hole_difficulty}$$

Shot difficulty was then split into three categorical buckets: easy (shots made around the 33rd percentile), medium (shots made around the 66th percentile), and hard (shots made around the 99th percentile). Observations regarding whether the putts were made were then put into these three buckets to calculate the make percentage. Using a logistic regression model, we were given a p-value that was below 0.001, which indicates a statistically significant relationship between the difficulty of the shot and the percentage of golfers who successfully made a putt.

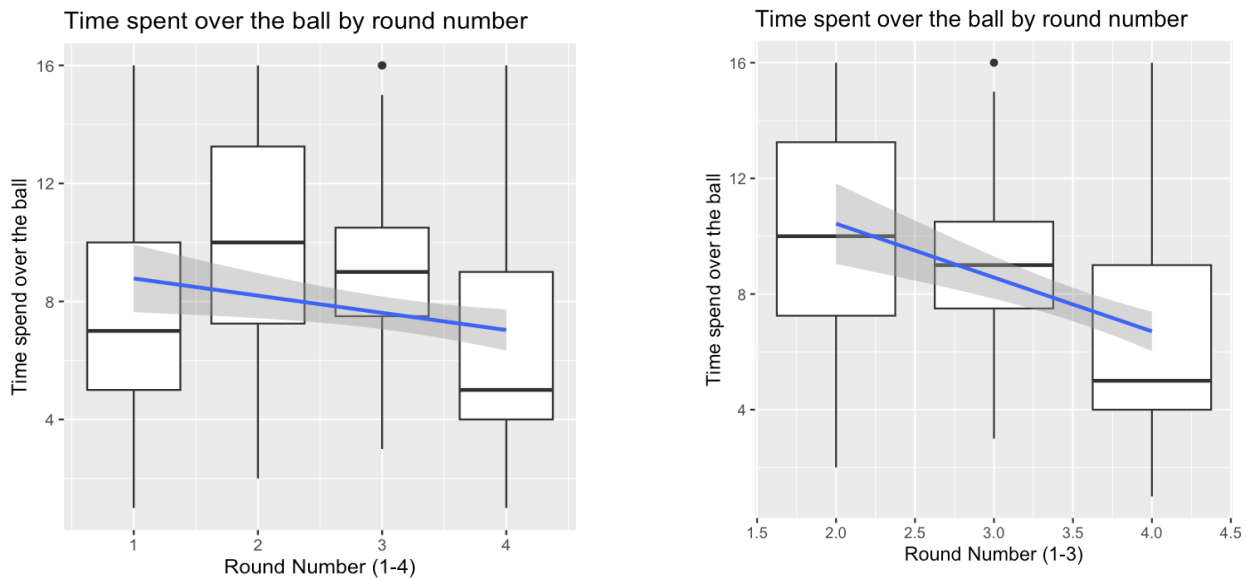


Figure 2 – Time spent over the ball by round number

Figure 2 shows the correlation between the time spent over the ball on each putt and the round number when the putt was played. The first graph shows that, while not obvious, there is a slight negative correlation between the two variables for the entire dataset. The correlation coefficient is -0.17, which is statistically significant at the 0.05 level. That being said, it is obvious that much of the correlation comes from the relationship for the latter 3 rounds. Removing the first round reveals a much stronger relationship, with a correlation coefficient of -0.34.

3. Insights from the Data

Insight 1:

Figure 1 reveals that shot difficulty has a much stronger influence on make percentage than distance alone would suggest. Although short putts are easier primarily because of their length, our shot difficulty metric shows that this assumption does not hold once course conditions are factored in. Easy and medium putts were both made at approximately 96%, while hard putts dropped sharply to 44%, even though many of those “hard” attempts occurred within the 0-6 foot range. This indicates that factors such as slope, break direction, and green speed can transform a short putt into a significantly more challenging one. The logistic regression further supports this interpretation. With a p-value below 0.001, the model confirms that the difficulty score is a statistically significant predictor of whether a putt is made. In practical terms, this means that golfers are not missing short putts simply because they misjudge distance. They are missing

because the various conditions of the course compound in ways that are not immediately obvious. Recognizing when a short putt is actually high-difficulty could help players adjust their course-reading strategy and decision-making, giving them an advantage that distance-based intuition alone would overlook.

Insight 2:

Figure 2 reveals an interesting relationship between the time spent over the ball for each putt and the round in which the putt took place. The negative correlation, which is statistically significant for all four rounds but is strengthened for the second through fourth rounds, is perhaps the opposite of what would be expected. It makes sense that, as the rounds progress and golfers are faced with tougher advancement situations, they are likely to be more careful with their putts and therefore spend more time with them. The data shows otherwise, however. There are many reasons why the first round might have a lower time spent over the ball; one possible explanation is that the first round is likely to include the most amateur or lower-ranking golfers. The average time spent in the first round is around 7 seconds. It's about 10 in the second round, 9 in the third, and 5 in the fourth.

It's hard to say why this relationship emerges in high-level golf. Likely, it's a build-up of confidence after playing several rounds on the same course. Whatever the reason, however, the negative correlation between round number and time spent over the ball for the putts could help golfers manage their time. Further investigation into this correlation for specific courses and even holes could give golfers and opponents insight into which holes they are the most confident in and how their approach evolves over the course of a tournament.

Critiques:

Looking at the data, there are several ways that we would be able to improve upon our work. A couple of the variables (Avg and Make %) that we collected gave any of our models or figures a chance of data leakage. These variables would cause this issue because they were taken from the very end of each season, meaning that some observations were getting data from the future if the tournament was played in the beginning or middle of the season. Overall, some of our variables contained many of the same results (more yes's than no's, or more 4s than other rounds). This could have led some of our figures to have skewed results and wouldn't be indicative of a general golf scenario. In the future, it would be a better idea to make sure that we get an even or close to even amount of certain results to avoid having possible skewed results.

In addition to some of the variables that could be removed or adjusted, there are several variables that we believe would add to the validity of our study if included in the future. Variables such as wind speed and surface condition (wet, dry, bumpy, etc) would allow us to better figure out what variables in the golfers' control would lead to better putting results. Variables such as previous

hole results could add insight into psychological pressure on top of the pressure situation variable that we had already recorded.

Because of the nature of the data collection that we did, there was some skew in the data. Because we had to rely on golf broadcasts available online — which often omit video data that we would need to collect some of our variables and show more of the later rounds — we were not able to evenly distribute the putting data over the rounds. Specifically, we ended up with more data from the fourth (final) rounds than from other rounds, which likely impacted the generalizability of our insights.

In the future, it would be a good idea and interesting to look at the factors that influence the score of the entire hole, rather than just the putt. This would allow us to compare and measure if some of the same factors (distance, hole difficulty, green speed) would also impact performance for the entire hole. In addition, this would give the hole difficulty and pressure situation in a different context to be put in as the opening swings have a crucial impact on the performance at the hole.